What makes an asset the dominant intermediary?

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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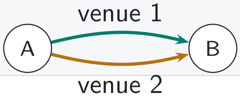
Transaction-level route structures

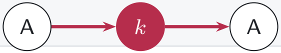
Direct swap
one transaction

Sequential vehicle route
one transaction

Parallel split
one transaction



Round trip
one transaction

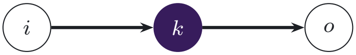


Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Measuring vehicle dominance

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



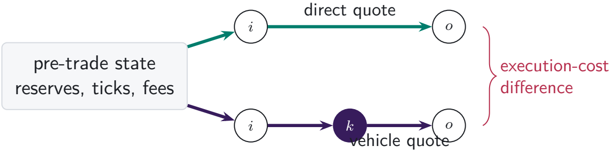
Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

The blockchain reveals chosen routes and contemporaneous alternatives

RESEARCH DESIGN

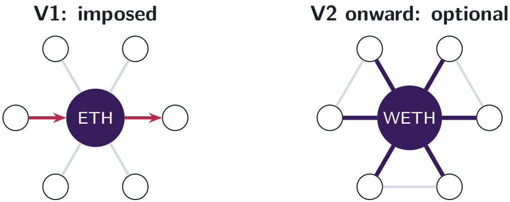
- FX data typically reveal executed trades.
- On-chain reserves and fees also let us price feasible alternatives.
- The comparison holds market conditions fixed at execution.



Protocol design first imposed ETH intermediation

INSTITUTIONAL FACT

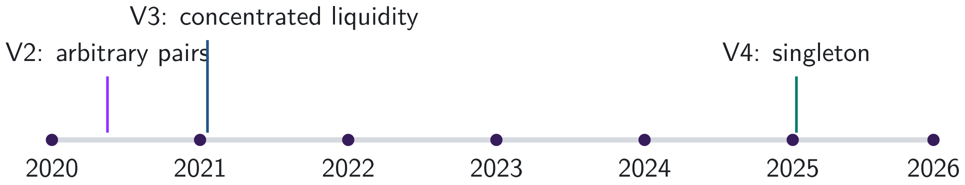
- Uniswap V1 created one exchange contract per token, paired with ETH.
- Token-to-token swaps therefore moved through ETH by construction.
- V2 allowed arbitrary pairs; the ETH-centred pairing network persisted.



ETH remained central after arbitrary pairs became possible.

The panel follows the market from V1 through V4

DATA



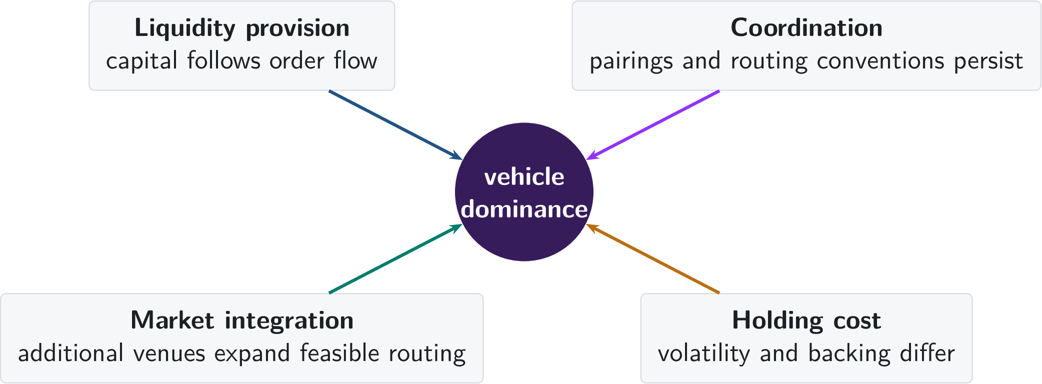
Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Four measures capture use, scale, and network position

1. **Vehicle share**
   Fraction of route units intermediated by asset k.
2. **Excess use**
   Intermediary share relative to endpoint demand.
3. **Network position**
   Degree, strength, and betweenness on the trading graph.
4. **Execution advantage**
   Direct-route output relative to a feasible vehicle route at the same pre-trade state.

Together, the measures distinguish use, scale, network position, and execution cost.

Four forces can sustain vehicle dominance



The empirical design separates formation, rotation, and persistence

FORMATION

Architecture changes
Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Panel variation
Count share, value share, excess use, network position, and route complexity over time.

PERSISTENCE

Contemporaneous alternatives
An incumbent retains volume despite a cheaper feasible route.

how the role begins

how its share moves

how long it survives

Vehicle dominance rotates toward stable numéraires

EXACT TWO-LEG ROUTES: EQUAL-WEIGHTED DAYS

16.9% $\longrightarrow$ 42.3%

+25.4 pp (SE 1.05 pp)

Holm $p = 3.36 \times 10^{-87}$; stable share within native + stable, 2024 to 2026.

STRICT-SUPPORT VALUE

32.7% $\longrightarrow$ 76.5%

+43.9 pp (SE 2.02 pp)

Holm $p = 4.48 \times 10^{-75}$; value supported within 20%.

Reading: a large late rotation, not a monotone time trend. Native use rebounds in 2023–24; stable value leads again from 2025-Q1, while count leadership is not yet sustained.

USDT is the transition margin

Object	2024	2026	Change
USDT count excess use	1.06	1.23	—
USDT strict-value excess use	0.59	1.42	—
Cross-venue count premium	paired-date change		+7.59 pp (0.88 pp)
Cross-venue strict-value premium	paired-date change		+12.00 pp (2.03 pp)
Intermediary − endpoint, strict value	−7.13 pp	+8.14 pp	+15.27 pp (1.50 pp)

92.1%

of the stable count-share increase comes from USDC + USDT.

Not a pooled "stablecoin" effect. USDC is already an excess-use vehicle and moves little; USDT crosses from below to above one on strict value. Backing designs remain separate mechanisms.

Integration expands; intermediation does not

CROSS-VENUE EXECUTION

2.0% ⟶ 57.2%

true intermediary routes, 2020 to 2026; 2026 strict-value share is 79.1%.

STABLE ROTATION IS STRONGER ACROSS VENUES

+7.45 pp (SE 1.33 pp)

TRUE INTERMEDIATION SHARE

	2022	2026	Δ
Full	13.95%	11.92%	-2.03 pp
Balanced	13.95%	8.60%	-5.35 pp

Calendar-HAC SEs: 0.39 pp (full), 0.38 pp (balanced).

Routing maturation changes reach, but it does not mechanically create more indirect trade. It remains a live rival: the final test conditions vehicle choice on fixed opportunity and search-efficiency cells.

Challenger displacement gradients are strong

30-day outcome	$\hat{\beta}$	SE	t
Challenger vehicle-share response	4.825	0.366	13.17
Incumbent vehicle-share response	-3.872	0.420	-9.22
Challenger overtakes incumbent	9.947	0.486	20.48

$N = 57,989$ pair-candidate observations; 2,213 date clusters. Exact alternating-projection absorption and CR1 rank correction.

What it says: the sparse challenger panel contains strong displacement gradients.

What it does not say: these gradients do not identify transition, routing efficiency, or persistence because the risk set does not yet condition on fixed feasible alternatives.

Architecture exposure lacks isolated substitution events

	5%	10%	25%
Sustained entries	162 (0)	139 (0)	78 (0)
Substitution exits	79 (0)	69 (0)	54 (0)
Role disappearances	7 (3)	7 (3)	7 (3)
Immediate entry contrast	–	–	–
Immediate exit contrast	–	–	–

WHAT THE COMPARISON ESTABLISHES

Zero clean substitution windows. Realised V4 share is endogenous, the pair-week centering is mechanical when shares sum to one, and role appearance/disappearance magnitudes are selected lifecycle endpoints. The design blocks an effect instead of manufacturing one.

The identifying comparison requires raw focal share on an ex ante feasible risk set, with exact-state reversals separating design response from calendar coincidence.

Three questions organize the evidence

1. **Formation**
   When does market architecture create or strengthen an intermediary role?
2. **Rotation**
   Which changes reflect intermediary demand, endpoint demand, trade size, or routing technology?
3. **Persistence**
   How long does incumbent intermediation survive when contemporaneous alternatives are available?

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

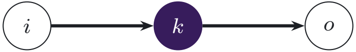
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

A2. Backing regimes vary over time

Native platform asset
ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised
DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

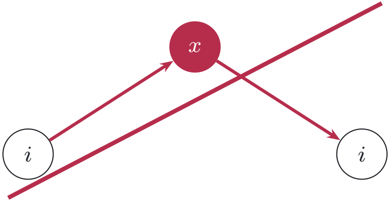
Tokenised external asset

WBTC, tokenised gold; wrapped claim on an external asset.

Heterogeneity tests use each asset's contemporaneous backing regime.

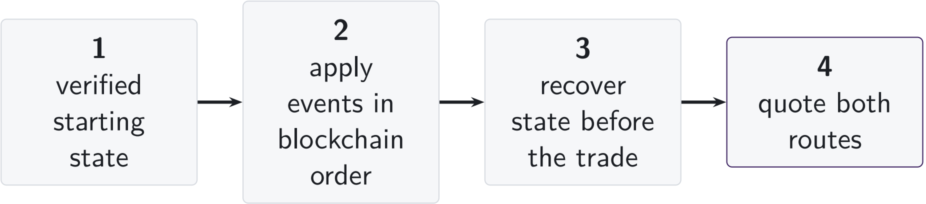
A3. Sample restrictions preserve value measurement

- Round trips are excluded from intermediation measures.
- Dollar values require consistent prices and verified token identities.
- We retain routes whose two legs imply dollar values within the pre-specified tolerance.



round trip: excluded from value-weighted vehicle measures

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. AMM designs require different state variables

Constant product Reserves, fee, and invariant.

Concentrated liquidity Active liquidity, price, tick, and initialised-tick changes.

StableSwap families Balances, amplification, rates, and oracle state.

Weighted pools Vault balances, weights, scaling, and rate providers.

V4 singleton Pool key, tick state, and fee for standard pools without custom hooks.

Estimates include protocol families whose balances, fees, and pricing state can be independently validated.

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.

same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions

1. **Primary panel**	day observations; date fixed effects
2. **Calendar sensitivity**	day observations; week and weekday fixed effects
3. **Aggregation robustness**	complete weeks; week fixed effects
4. **Transition series**	exact 1-, 7-, 30-, and 120-day links; HAC inference

Weekly ratios are formed from weekly sums; results are reported for weeks ending on each day of the week.

A8. Count and value define separate dominance margins

Count-weighted share

$$S^N_{k,t} = \frac{N^I_{k,t}}{\sum_j N^I_{j,t}}$$

Each reconstructed route carries one unit of weight.

Value-weighted share

$$S^V_{k,t} = \frac{\mathrm{Vol}^I_{k,t}}{\sum_j \mathrm{Vol}^I_{j,t}}$$

Each route carries reliably priced dollar value on common support.

Differences between count and value shares may reflect trade size, route structure, or the mix of intermediary assets.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$S^I_{k,t} = \frac{N^I_{k,t}}{\sum_j N^I_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$S^E_{k,t} = \frac{N^{\text{in}}_{k,t} + N^{\text{out}}_{k,t}}{\sum_j N^{\text{in}}_{j,t} + N^{\text{out}}_{j,t}}$$

How often asset k is what traders buy or sell.

Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are distinct quantities

Deposited capital
Funds committed by liquidity providers, established from independently valued holdings or audited claims. **Inventory**

Exact token amounts held by the pool.

Executable depth
Amount that can be traded for a given price impact under current pool state.
Quote quality

Execution price for trade size q along feasible routes.

Capital enters only where holdings and ownership are independently validated; execution depth comes from the pool's pricing function.

A11. Integration changes both reach and route choice

Available trading opportunities

- active venues
- feasible paths
- available intermediary assets

Route choice conditional on availability

- direct versus intermediated
- single- versus cross-venue
- sequential versus split routing

We test whether stablecoin intermediation rises more when additional venues and routes are available, conditional on endpoint pair and date.

A12. References: vehicle currencies and market structure

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